

1. Solve.

$$7^{-x+4} = 17$$

$x \approx$

(Simplify your answer. Type an integer or decimal rounded to four decimal places as needed.)

2. A minute ad during a campaign in 1973 cost \$200,000. The price tag for a 30-second ad slot during a 2005 campaign was \$3 million. The cost of a 30-second slot of advertising for a campaign can be modeled by $y = 0.0000963(1.104^x)$, where x is the number of years after 1900 and y is the cost in millions.

Complete (a) and (b) below.

a. According to the model, what was the cost of a 30-second campaign ad in 2000?

The cost of a 30-second campaign ad in 2000 was \$.

(Simplify your answer. Round to the nearest dollar as needed.)

b. The cost of a 30-second campaign ad in 2015 was \$7.0 million. Does the model underestimate or overestimate the cost of these ads?

The model (1) _____ the cost for a 30-second campaign ad. The model estimates the cost for 2015 at \$.

(Simplify your answer. Round to the nearest dollar as needed.)

- (1) ☐ overestimates
☐ underestimates

3. Solve the logarithmic equation.

$$4 + 2 \ln x = 10$$

$x =$

(Simplify your answer. Type an exact answer, using e as needed.)